

ARES Protocol — Use Cases

The same battery, real commands: JSON, XML, and mixed

One protocol, several encodings — every command shown in full. The register / heartbeat / claim / release battery runs over **REST/JSON** (`curl`), over **REST/XML** (`curl`), and **mixed** across both, against one coordinator and one 7-zone workspace (`dock_a=1 ... dock_c=3, zone_4=4 ... zone_7=7`). Zones 1–4 and 6 are exclusive; zones 5 and 7 are shared with capacity 2. Server version `0.1.3`. Every reply is captured from a live run.

Auto-release: a robot loses its zones when it stops heart-beating — no manual reset. Register with a high `<alive>` (e.g. `9999`) to avoid constant heartbeats; the server reclaims a zone only after `2×` the `alive` interval passes with no heartbeat (section F uses a short `alive` to show it).

Leases: `<lease_time>` is in **seconds of wall clock**. A claim carrying one is dropped that many seconds after it is granted, whether or not the robot keeps heart-beating; omitting it (or sending `0`) holds the zone until it is released or the robot goes quiet.

Reason codes (after `0` ok / `1` mismatched key): register `2` already registered, `3` bad id, `5` not permitted (id not provisioned, or keyless registration disabled), `6` fleet full. Heartbeat `2` not registered. Claim `2` conflict, `3` capacity, `4` unknown id, `5` bad request. Release `2` no such lease.

1 JSON (REST) — `curl`

Set `B=http://<host>:8080/ares/v1`. JSON is the default wire format: the `content-type` header below is explicit for clarity, but a body with no `content-type` is read as JSON and a request with no `Accept` is answered in JSON. Replies shown after `# ->`.

1.1 A. Read (no key)

```
curl -s "$B/health"
# -> {"status":"ok","version":"0.1.3","datum":{"lat":52.0,"lon":5.0,"alt":0.0}}
curl -s "$B/zones" # -> [ ... 8 zones: fixed_yard, dock_a ... dock_c, zone_4 ... zone_7 ]
curl -s "$B/zones/3"
# -> {"id":"...0103","numeric_id":3,"name":"dock_c","kind":"zone",...
#     "properties":{"traffic.policy":"exclusive",...}}
curl -s "$B/zones/99" # -> {"message":"unknown zone id Numeric(99)"}
curl -s "$B/fleet/snapshot"
# -> {"robots":[],"requests":[],"leases":[],"inactive_robot_ids":[],"events":[],...}
```

1.2 B. Register

```
curl -s -X POST "$B/robots" -H 'content-type: application/json' \
  -d '{"robot":"91","key":"1234","alive":9999}'
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/robots" -H 'content-type: application/json' \
  -d '{"robot":"91","key":"1234"}' # again -> already registered
# -> {"decision":0,"reason":2}
curl -s -X POST "$B/robots" -H 'content-type: application/json' \
  -d '{"robot":"92"}' # no key -> default
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/robots" -H 'content-type: application/json' \
  -d '{"robot":"abc"}' # bad id
# -> {"decision":0,"reason":3}
```

1.3 C. Heartbeat

```
curl -s -X POST "$B/robots/91/heartbeat" -H 'content-type: application/json' \
  -d '{"key":"1234","zone":3}' # ok
# -> {"decision":1,"reason":0}
```

```

curl -s -X POST "$B/robots/91/heartbeat" -H 'content-type: application/json' \
  -d '{"key":"9999","zone":3}' # wrong key
# -> {"decision":0,"reason":1}
curl -s -X POST "$B/robots/999/heartbeat" -H 'content-type: application/json' \
  -d '{"zone":3}' # not registered
# -> {"decision":0,"reason":2}
curl -s -X POST "$B/robots/91/heartbeat" -H 'content-type: application/json' \
  -d '{"key":"1234","zone":-1}' # location unknown
# -> {"decision":1,"reason":0}

```

1.4 D. Claim

Setup: `curl -s -X POST "$B/robots" -H 'content-type: application/json' -d '{"robot":"94","key":"4444","alive":9999}'`.

`id` is an **array** even for a single resource — `[5]`, not `5`.

```

curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"key":"1234","robot":"91","id":[5]}' # granted
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"robot":"92","id":[5]}' # 5 held by 91
# -> {"decision":0,"reason":2,"blocked":5}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"robot":"92","id":[6,5]}' # atomic; 5 held
# -> {"decision":0,"reason":2,"blocked":5}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"robot":"92","id":[6]}' # granted
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"key":"4444","robot":"94","id":[7],"access_mode":1,"lease_time":30}'
# -> {"decision":1,"reason":0} # exclusive + lease
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"key":"1234","robot":"91","id":[1],"access_mode":2}' # access_mode 2 = shared
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"key":"1234","robot":"91","id":[2],"access_mode":3}' # 3+ reserved
# -> {"decision":0,"reason":5}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"key":"1234","robot":"91","id":[99]}' # unknown zone
# -> {"decision":0,"reason":4,"blocked":99}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"key":"0000","robot":"91","id":[2]}' # wrong key
# -> {"decision":0,"reason":1}

```

1.5 E. Release

`id` here is a **single** value, not an array — one resource is released at a time.

```

curl -s -X POST "$B/leases/release/zone" -H 'content-type: application/json' \
  -d '{"key":"1234","robot":"91","id":5}' # released
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/leases/release/zone" -H 'content-type: application/json' \
  -d '{"key":"1234","robot":"91","id":5}' # nothing held now
# -> {"decision":0,"reason":2}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
  -d '{"robot":"92","id":[5]}' # now free -> granted
# -> {"decision":1,"reason":0}

```

1.6 F. Auto-release on heartbeat timeout

```

curl -s -X POST "$B/robots" -H 'content-type: application/json' \
  -d '{"robot":"95","key":"5","alive":1}' # short alive
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \

```

```

-d '{"key":"5","robot":"95","id":[3]}' # 95 takes zone 3
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/robots" -H 'content-type: application/json' \
-d '{"robot":"96","key":"6"}'
# -> {"decision":1,"reason":0}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
-d '{"key":"6","robot":"96","id":[3]}' # 95 still holds it
# -> {"decision":0,"reason":2,"blocked":3}
sleep 3 # 95 (alive 1) misses 2x its interval -> inactive, zone 3 auto-released
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
-d '{"key":"6","robot":"96","id":[3]}' # now granted
# -> {"decision":1,"reason":0}

```

Same protocol, different clothes. Every decision and reason above is identical to the XML chapter that follows — only the encoding differs. Pick a format per client; the coordinator does not care which one a robot speaks.

2 XML (REST) — `curl`

Set `B=http://<host>:8080/ares/v1`. Send `content-type: application/xml` and, for the read endpoints, ask for `Accept: application/xml`. Replies shown after `# ->`.

2.1 A. Read (no key)

```

curl -s "$B/health"
# -> {"status":"ok","version":"0.1.3"}
curl -s "$B/zones" # -> [ ... zones: dock_a, dock_b, dock_c, zone_4 ... zone_7 ]
curl -s "$B/zones/3" # -> {"numeric_id":3,"name":"dock_c",...,"traffic.policy":"exclusive"}
curl -s "$B/zones/99" # -> {"message":"unknown zone id Numeric(99)"}
curl -s "$B/fleet/snapshot" # -> {"robots":[],"requests":[],"leases":[]}

```

2.2 B. Register

```

curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
-d '<reg><robot>91</robot><key>1234</key><alive>9999</alive></reg>'
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
-d '<reg><robot>91</robot><key>1234</key></reg>' # again -> already registered
# -> <reply><decision>0</decision><reason>2</reason></reply>
curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
-d '<reg><robot>92</robot></reg>' # no key -> default
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
-d '<reg><robot>abc</robot></reg>' # bad id
# -> <reply><decision>0</decision><reason>3</reason></reply>

```

2.3 C. Heartbeat

```

curl -s -X POST "$B/robots/91/heartbeat" -H 'content-type: application/xml' \
-d '<hb><key>1234</key><zone>3</zone></hb>' # ok
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/robots/91/heartbeat" -H 'content-type: application/xml' \
-d '<hb><key>9999</key><zone>3</zone></hb>' # wrong key
# -> <reply><decision>0</decision><reason>1</reason></reply>
curl -s -X POST "$B/robots/999/heartbeat" -H 'content-type: application/xml' \
-d '<hb><zone>3</zone></hb>' # not registered
# -> <reply><decision>0</decision><reason>2</reason></reply>
curl -s -X POST "$B/robots/91/heartbeat" -H 'content-type: application/xml' \
-d '<hb><key>1234</key><zone>-1</zone></hb>' # location unknown
# -> <reply><decision>1</decision><reason>0</reason></reply>

```

2.4 D. Claim

Setup: `curl -s -X POST "$B/robots" -H 'content-type: application/xml' -d '<reg><robot>94</robot><key>4444</key><alive>9999</alive></reg>'`

```
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>1234</key><robot>91</robot><id>5</id></claim>' # granted
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><robot>92</robot><id>5</id></claim>' # 5 held by 91
# -> <reply><decision>0</decision><reason>2</reason><blocked>5</blocked></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><robot>92</robot><id>6</id><id>5</id></claim>' # atomic; 5 held
# -> <reply><decision>0</decision><reason>2</reason><blocked>5</blocked></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><robot>92</robot><id>6</id></claim>' # granted
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>4444</key><robot>94</robot><id>7</id>
    <access_mode>1</access_mode><lease_time>30</lease_time></claim>' # exclusive + lease
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>1234</key><robot>91</robot><id>1</id>
    <access_mode>2</access_mode></claim>' # access_mode 2 = shared
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>1234</key><robot>91</robot><id>2</id>
    <access_mode>3</access_mode></claim>' # 3+ reserved
# -> <reply><decision>0</decision><reason>5</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>1234</key><robot>91</robot><id>99</id></claim>' # unknown zone
# -> <reply><decision>0</decision><reason>4</reason><blocked>99</blocked></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>0000</key><robot>91</robot><id>2</id></claim>' # wrong key
# -> <reply><decision>0</decision><reason>1</reason></reply>
```

2.5 E. Release

```
curl -s -X POST "$B/leases/release/zone" -H 'content-type: application/xml' \
  -d '<rel><key>1234</key><robot>91</robot><id>5</id></rel>' # released
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/leases/release/zone" -H 'content-type: application/xml' \
  -d '<rel><key>1234</key><robot>91</robot><id>5</id></rel>' # nothing held now
# -> <reply><decision>0</decision><reason>2</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><robot>92</robot><id>5</id></claim>' # now free -> granted
# -> <reply><decision>1</decision><reason>0</reason></reply>
```

2.6 F. Auto-release on heartbeat timeout

```
curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
  -d '<reg><robot>95</robot><key>5</key><alive>1</alive></reg>' # short alive
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>5</key><robot>95</robot><id>3</id></claim>' # 95 takes zone 3
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
  -d '<reg><robot>96</robot><key>6</key></reg>'
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
  -d '<claim><key>6</key><robot>96</robot><id>3</id></claim>' # 95 still holds it
# -> <reply><decision>0</decision><reason>2</reason><blocked>3</blocked></reply>
sleep 3 # 95 (alive 1) misses 2x its interval -> inactive, zone 3 auto-released
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
```

```
-d '<claim><key>6</key><robot>96</robot><id>3</id></claim>' # now granted
# -> <reply><decision>1</decision><reason>0</reason></reply>
```

access_mode 2 now means **shared**, not “reserved” — on a free zone it is granted, and only lets robots coexist on a **shared** -policy zone with capacity > 1. On a free exclusive zone it behaves like a normal claim. **access_mode 3**+ is still reserved (→ reason **5**).

3 Mixed (REST/JSON + REST/XML, one coordinator)

Two robots on **different encodings** against the **same** server: robot **91** speaks REST/XML (**curl**) and robot **93** speaks REST/JSON (**curl**). We register one on each, then run the experiments — a claim on either blocks the other, in both directions, and a release on one frees the zone for the other.

```
# --- register: 91 over XML, 93 over JSON ---
curl -s -X POST "$B/robots" -H 'content-type: application/xml' \
-d '<reg><robot>91</robot><key>1234</key><alive>9999</alive></reg>'
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/robots" -H 'content-type: application/json' \
-d '{"robot": "93", "key": "3333", "alive": 9999}'
# -> {"decision": 1, "reason": 0}

# --- experiment 1: XML robot 91 takes zone 4, JSON robot 93 is blocked ---
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
-d '<claim><key>1234</key><robot>91</robot><id>4</id></claim>'
# -> <reply><decision>1</decision><reason>0</reason></reply> (91 holds zone 4)
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
-d '{"key": "3333", "robot": "93", "id": [4]}'
# -> {"decision": 0, "reason": 2, "blocked": 4} (blocked by XML 91)

# --- experiment 2: JSON robot 93 takes zone 6, XML robot 91 is blocked ---
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
-d '{"key": "3333", "robot": "93", "id": [6]}'
# -> {"decision": 1, "reason": 0} (93 holds zone 6)
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
-d '<claim><key>1234</key><robot>91</robot><id>6</id></claim>'
# -> <reply><decision>0</decision><reason>2</reason><blocked>6</blocked></reply> (blocked by JSON 93)

# --- experiment 3: XML robot 91 releases zone 4, JSON robot 93 can now take it ---
curl -s -X POST "$B/leases/release/zone" -H 'content-type: application/xml' \
-d '<rel><key>1234</key><robot>91</robot><id>4</id></rel>'
# -> <reply><decision>1</decision><reason>0</reason></reply>
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
-d '{"key": "3333", "robot": "93", "id": [4]}'
# -> {"decision": 1, "reason": 0} (now free -> granted)

# --- experiment 4: JSON robot 93 takes zone 2, XML robot 91 is blocked ---
curl -s -X POST "$B/claims/zone" -H 'content-type: application/json' \
-d '{"key": "3333", "robot": "93", "id": [2]}'
# -> {"decision": 1, "reason": 0} (93 holds zone 2)
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
-d '<claim><key>1234</key><robot>91</robot><id>2</id></claim>'
# -> <reply><decision>0</decision><reason>2</reason><blocked>2</blocked></reply> (blocked by JSON 93)

# --- experiment 5: JSON robot 93 releases zone 2, XML robot 91 takes it ---
curl -s -X POST "$B/leases/release/zone" -H 'content-type: application/json' \
-d '{"key": "3333", "robot": "93", "id": [2]}'
# -> {"decision": 1, "reason": 0}
curl -s -X POST "$B/claims/zone" -H 'content-type: application/xml' \
-d '<claim><key>1234</key><robot>91</robot><id>2</id></claim>'
# -> <reply><decision>1</decision><reason>0</reason></reply> (now free -> granted)
```

Captured live: both encodings via `curl` against one coordinator on the same seven-zone workspace. The encoding is a client's own choice — the coordinator sees one fleet.